

Structured records

CSV / JSON

Structured extractor

deterministic schema mapping
fixed high confidence

Unstructured text

plain text

LLM extractor

Llama-3.1-8B-Instruct (OpenRouter)
self-reported confidence 0.80-0.96

Probabilistic knowledge base

observations
conservative Noisy-OR + conflict penalty
(DySECT Eqs. 1-2; $\lambda=0.75$, $\tau=0.88$)
per-observation provenance

Feedback controller

missing facts / false patterns /
under-covered predicates

Gold

labels,
training
split only

QLoRA fine-tuning

Mistral-7B-Instruct-v0.3
run manually on a rented GPU

Provenance filter

keeps admitted triples with structured
corroboration; builds synthetic examples

adapter (closed-loop
test only)

training
data

feedback hint
+ locked context

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